

Program-5

Implement and demonstrate the FIND S algorithm for finding the most specific hypothesis based on a given set of tanning data sample . read the tanning data from a .csv file

Aim

The aim of this experiment is to implement and demonstrate the FIND-S algorithm to determine the most specific hypothesis that fits a given set of training data samples by reading the dataset from a CSV file and analyzing only the positive examples.

Theory

The FIND-S (Find-Specific) algorithm is a fundamental concept learning algorithm in machine learning used to determine the most specific hypothesis that is consistent with a given set of training data. It is a supervised learning algorithm that works on labeled data and focuses only on positive training examples while ignoring negative examples. The algorithm begins with the most specific hypothesis, typically represented by null or '0' values, and iteratively updates it as it processes each positive example. For every attribute in the training instance, if the current hypothesis value differs from the example, it is generalized by replacing it with a wildcard symbol ('?'), which indicates that any value is acceptable for that attribute. This process continues until all positive examples are covered. The final hypothesis produced by the FIND-S algorithm represents the most specific generalization that is consistent with all positive training instances. Although the algorithm is simple and easy to implement, it has certain limitations such as ignoring negative examples, inability to handle noisy data, and assuming that the training data is perfectly consistent.

Result

The FIND-S algorithm was successfully implemented using training data from a CSV file. The algorithm processed only the positive examples and gradually generalized the hypothesis by replacing differing attribute values with wildcard symbols. The final hypothesis obtained represents the most specific generalization that is consistent with all positive training samples.

```
[1]: import csv

# Load data from CSV file
with open('training_data.csv', 'r') as file:
    data = list(csv.reader(file))

# Separate header and data
header = data[0]
data = data[1:]

# Initialize hypothesis with most specific values
hypothesis = ['0'] * (len(header) - 1)

print("Initial Hypothesis:", hypothesis)

# Apply FIND-S Algorithm
for row in data:
    if row[-1].lower() == 'yes': # consider only positive examples
        for i in range(len(hypothesis)):
            if hypothesis[i] == '0':
                hypothesis[i] = row[i]
            elif hypothesis[i] != row[i]:
                hypothesis[i] = '?'

        print("Updated Hypothesis:", hypothesis)

print("\nFinal Hypothesis:", hypothesis)

Initial Hypothesis: ['0', '0', '0', '0', '0', '0']
Updated Hypothesis: ['Sunny', 'Warm', 'Normal', 'Strong', 'Warm', 'Same']
Updated Hypothesis: ['Sunny', 'Warm', '?', 'Strong', 'Warm', 'Same']
Updated Hypothesis: ['Sunny', 'Warm', '?', 'Strong', 'Warm', 'Same']
Updated Hypothesis: ['Sunny', 'Warm', '?', 'Strong', '?', '?']
Updated Hypothesis: ['?', 'Warm', '?', '?', '?', '?']
Updated Hypothesis: ['?', 'Warm', '?', '?', '?', '?']

Final Hypothesis: ['?', 'Warm', '?', '?', '?', '?']
```